

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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Anonymous submission

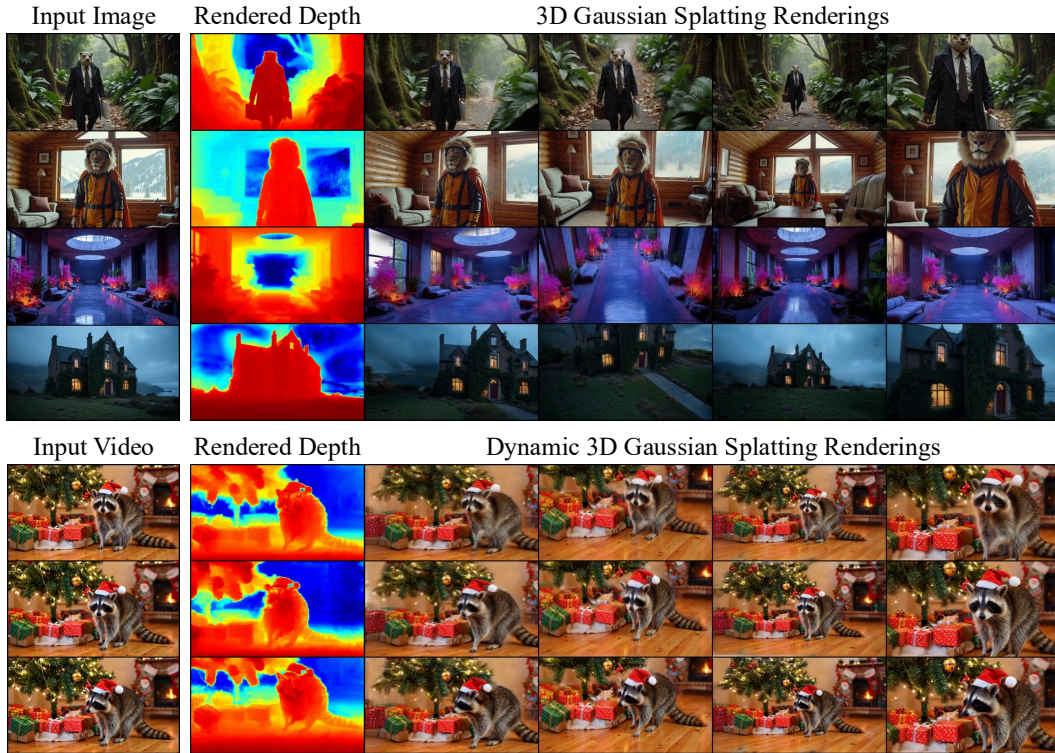


Figure 1: **Feed-forward 3D and 4D scene generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a single feed-forward pass by self-distilling a camera-controlled video diffusion model, using no real-world multi-view data. From a video input (bottom), *Lyra* infers a dynamic 3DGS that affords independent control over both time (rows) and viewpoint (columns).

ABSTRACT

The capacity to synthesize virtual environments underpins a broad spectrum of applications, from gaming and content creation to physical-AI domains such as robotics, autonomous driving, and industrial AI. Learning-based 3D reconstruction, however, remains fundamentally constrained by its reliance on captured real-world multi-view data—supervision that is costly to acquire and rarely available at the scale and diversity these applications demand. Video diffusion models offer a compelling route around this bottleneck: trained on internet-scale corpora, they exhibit remarkable generative imagination and broad coverage, yet their purely two-dimensional output withholds the explicit geometry required whenever an agent must navigate and physically interact with its surroundings. We resolve this tension with a self-distillation framework that transfers the 3D knowledge implicitly encoded in a camera-controlled video diffusion model into an explicit 3D Gaussian Splatting (3DGS) representation, thereby dispensing with multi-view training data altogether. Concretely, we augment the video model’s standard RGB decoder with a parallel 3DGS decoder that operates in the same latent space and

is supervised directly by the RGB decoder’s output, so that the student decoder is trained end-to-end from supervision the video diffusion model synthesizes for itself. At inference, the model produces an explicit 3D scene from either a text prompt or a single image and renders it in real time, with no per-scene optimization and no post-processing; the same framework extends, with minimal modification, to dynamic 3D scene generation from a monocular input video. Empirically, the framework attains state-of-the-art single-image-to-3D reconstruction across three standard benchmarks and, in the previously unaddressed feed-forward 4D setting, substantially outperforms a strong feed-forward baseline in single-video 4D generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics, and a prerequisite for applications spanning film-making, VR/AR, and physical AI with closed-loop simulation. These applications share a single, exacting requirement: an explicit 3D representation that simultaneously supports real-time rendering, physical interaction, and multi-view-consistent synthesis.

Neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recovers such representations from posed images, but its reliance on accurate camera poses and high-quality captures sharply limits scalability. The difficulty compounds for dynamic scenes, which typically demand synchronized multi-camera rigs (Li et al., 2022), and, more fundamentally, reconstruction is confined to the observed content and cannot extrapolate beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) were introduced to relax the per-scene optimization requirement, yet they inherit a bottleneck of their own: the scarcity of diverse, large-scale 3D training data, which curtails out-of-domain generalization. These constraints, disparate as they appear, trace back to a single root cause—the dependence on captured 3D or multi-view data.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a compelling escape from this data bottleneck. Trained on massive internet-scale corpora, they attain high fidelity and broad generalization across diverse environments; and because they observe real-world videos under varied camera trajectories—handheld motion, cinematic panning, drone footage—they implicitly encode rich cues about the underlying 3D world without ever consuming multi-view training data. Unlike reconstruction approaches (Charatan et al., 2024), these generative models can moreover hallucinate plausible content beyond what the input frames reveal. The obstacle is that they emit only 2D frames and expose no explicit 3D representation, which precludes their direct use in simulation and in any downstream task that requires geometric consistency, long-horizon coherence, and physical interaction. Crucially, recent work (Wang et al., 2024e; Ren et al., 2025) shows that video models can be endowed with explicit camera control using only relatively small-scale posed datasets. This capacity to synthesize *posed* image sequences is precisely what renders such a model usable as both the input to and the supervision for a 3D reconstruction model—and it is the lever our method exploits.

Motivated by this observation, we unify the two paradigms—reconstruction and generation—into a single formulation we term *generative 3D scene reconstruction*, and we realize it in *Lyra*, a framework that decodes explicit 3D environments directly from the latent representations of a video diffusion model in a single forward pass. At its core (Fig. 2) is a self-distillation framework in which a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a camera-controlled video diffusion model. Given a single input image or video, we sample a camera trajectory as the conditioning signal, denoise the resulting video latent, and decode it along two parallel branches: the pre-trained RGB decoder synthesizes a video sequence, while the 3DGS decoder produces an explicit 3D representation. The two branches close a self-distillation loop in which the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we sample multiple camera trajectories, generate the corresponding video latents, and train the 3DGS decoder to fuse information across them while resolving long-horizon and multi-view inconsistencies.

This formulation confers three decisive advantages. (i) It sources large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, obviating real-world multi-view capture entirely. (ii) By operating in the latent space of the video model rather than in pixel space, it aggregates many views efficiently and circumvents the memory overhead that constrains pixel-

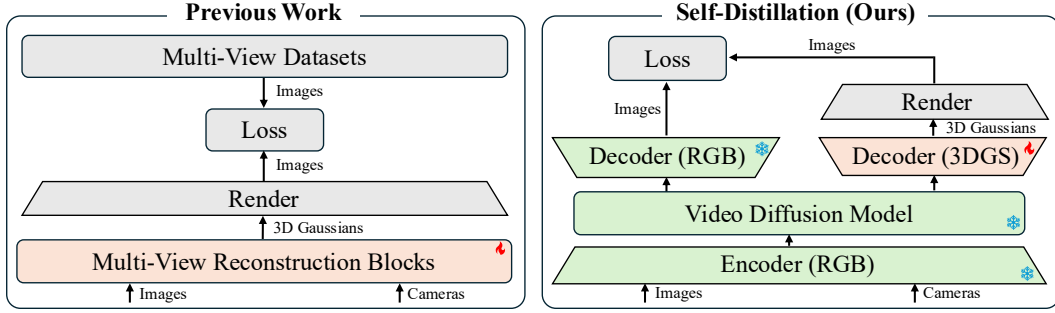


Figure 2: **Self-distillation framework of *Lyra*.** Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). We instead propose a self-distillation framework (right) for generative 3D scene reconstruction, in which a pre-trained camera-controlled video diffusion model supervises the renderings of the 3DGS decoder (student) through its RGB decoder output (teacher). Because the video model is pre-trained on diverse 2D video, it supplies correspondingly diverse multi-view supervision—the source of the generalization we observe.

space feed-forward reconstruction: it fuses the 726 views produced by six trajectories, more than an order of magnitude beyond the handful of frames such methods can accommodate. (iii) Its explicit 3DGS output enforces geometric consistency by construction, yielding representations that render in real time and are directly usable in downstream tasks such as physical simulation. Consequently, at inference *Lyra* generates high-quality 3D Gaussian scenes from monocular input, *without any additional optimization or post-processing*.

The same self-distillation framework extends, with minimal modification, to dynamic 4D generation from monocular video: the video model (teacher) now supplies space–time supervision, and the student learns to produce time-conditioned 3DGS representations that support novel-view synthesis of dynamic scenes—a feed-forward 4D generation setting previously unaddressed.

In summary, this work makes the following contributions:

- **Generative 3D scene reconstruction via self-distillation.** We introduce *generative 3D scene reconstruction* and realize it through a self-distillation framework that trains a 3DGS student decoder under the supervision of a pre-trained camera-controlled video diffusion model, eliminating the need for any captured multi-view real-world data.
- **Latent-space fusion and a 4D extension.** Because the student decoder operates directly in the video model’s latent space, it aggregates hundreds of generated views in a single feed-forward pass—far beyond the capacity of pixel-space reconstruction—and the framework extends naturally to 4D reconstruction from a monocular video, a setting not previously addressed by feed-forward methods.
- **State-of-the-art static reconstruction and a strong 4D result.** The resulting model applies to diverse static and dynamic scenes, attaining state-of-the-art single-image-to-3D reconstruction on three standard benchmarks and substantially outperforming a strong feed-forward baseline in single-video 4D generation.

2 RELATED WORK

Multi-view image generation. Early work on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) centered on object-centric scenes without background; subsequent efforts (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) scaled these generators to the scene level. CAT3D (Gao et al., 2024b) repurposes a pre-trained image diffusion model to synthesize multi-view images from any generated or real image, after which a separate stage reconstructs those images into an explicit representation such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). The paradigm has since been extended to dynamic 3D scenes rendered from multiple viewpoints via space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Impressive as these results are, grounding the generations in an explicit 3D representation hinges on a per-scene optimization stage whose cost is not amortized across scenes. We dispense with this two-stage recipe: rather than first synthesizing multi-view images and then reconstructing them by optimization, we decode a 3D scene *directly* from text or a single image.

Camera-conditioned video generation. Fine-tuning video diffusion models for 3D camera control has advanced rapidly. MotionCtrl (Wang et al., 2024e) pioneered the direction by conditioning pre-trained video models on camera poses; follow-up work (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) encodes cameras as Plücker coordinates for pixel-wise conditioning, while a parallel line (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) pursues training-free camera control. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) attained compelling camera-controlled video synthesis by leveraging large-scale synthetic training data. Effective as they are, these methods produce video that is not grounded in 3D. Our work is complementary and takes up precisely where they leave off: we address feed-forward 3D generation from camera-controlled video models, enabling novel-viewpoint rendering *after* generation. Concretely, we build on GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and employ it as the teacher within our self-distillation framework.

Feed-forward 3D models. Early feed-forward 3D reconstruction likewise targeted object-centric scenes without background, directly predicting a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from text or an image. More recent methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) address scene-level reconstruction, typically regressing per-pixel 3D Gaussians from one or more input images. Their persistent limitation is generalization: most scene-level reconstructors are tied to their training distribution—e.g., RealEstate10K (Zhou et al., 2018)—and transfer poorly to generated scenes. A recent thread therefore extends feed-forward reconstructors toward scene *generation*. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to emit pointmaps alongside multi-view images for feed-forward reconstruction, and, most closely related to us, Wonderland (Liang et al., 2025a) drives a feed-forward network with a camera-controlled video model to produce 3D Gaussians. We depart from both in one fundamental respect: we cast the camera-controlled video model as a *teacher* within a self-distillation framework and train the student 3D model *without any real-world multi-view data*. Just as consequentially, the framework admits a straightforward extension to feed-forward 4D scene generation, a task not previously addressed.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our central idea is to distill the 3D knowledge implicitly embedded in a video diffusion model into an explicit 3DGS decoder that produces high-quality 3D representations. To this end, we formalize the problem as a teacher–student framework: the video diffusion model (teacher) generates RGB videos that supervise a 3DGS decoder (student) operating in the same latent space, as illustrated in Fig. 2. This section develops the framework in four steps. We first describe the video diffusion backbone (Sec. 3.1) and the role it plays as the teacher during self-distillation of the 3DGS decoder (Sec. 3.2). We then motivate and specify the design of the 3DGS decoder (Sec. 4) and, finally, detail the minimal modifications that adapt the pipeline to dynamic 3D scenes (Sec. 5).

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised exclusively by the camera-controlled video diffusion model, the 3D consistency of that model is the foundation on which everything else rests. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model selected for precisely this property, and summarize below the design choices that make it a reliable teacher.

Spatiotemporal 3D cache. To enforce temporal consistency and precise camera control, GEN3C maintains a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ derived from the input image(s) or video. Each element $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the estimated depth (Wang et al., 2024a) of an RGB image observed from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the temporal length (number of frames) and V the number of camera views.

Rendering and structured guidance. To exploit this cache, GEN3C rasterizes each point cloud from arbitrary target camera poses, yielding RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering operator that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D image plane under the camera pose $\mathbf{C}^t$ at time t . Given a pose sequence $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements yields V videos of length L , which stack into per-view image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$. The disocclusion

masks delineate the regions the video diffusion model must inpaint; together, the renderings and masks constitute structured visual guidance that conditions the subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generators conventionally employ a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space, rendering both training and inference tractable. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ maps it to the latent code $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$, and the entire diffusion process is carried out within this latent space. The output video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is recovered by the pre-trained VAE decoder $\mathcal{D}_{rgb}$. We instantiate this backbone with the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model, for which the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal and spatial compression factors $\tau = 8$ and $\sigma = 8$, respectively.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as the student in a teacher–student paradigm in which the camera-controlled video diffusion model $\mathcal{V}$ acts as the teacher. To secure broad and varied supervision, we curate a large-scale collection of diverse text prompts with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and expand each image into multi-view sequences with GEN3C (Ren et al., 2025). Because this supervision is synthesized entirely on the fly, it scales with compute alone and requires no real-world capture.

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. This single latent is decoded along two parallel branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ recovers video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ emits explicit 3D Gaussians $\mathbf{G}$. The student’s rendered views $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ are supervised to match the teacher’s RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$, closing the self-distillation loop. Because both branches decode the identical latent, the teacher provides pixel-aligned supervision for every view the student renders.

Multi-trajectory supervision. To maximize the viewpoint coverage available for a given input image, we sample $V = 6$ camera trajectories per image, as illustrated in Fig. 3. For each trajectory v we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Propagating these caches through the video diffusion model $\mathcal{V}$ produces per-trajectory latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ then learns to fuse the multiple latents $\mathbf{z}^v$ into a single coherent set of Gaussians $\mathbf{G}$ while inpainting the disoccluded regions. This design confers two complementary benefits: operating in latent space lets the decoder aggregate many trajectories under a modest memory budget, and the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency while directly supporting real-time rendering and downstream simulation—all from supervision synthesized entirely by $\mathcal{V}$, with no real-world capture.

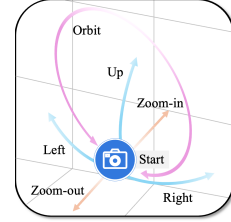


Figure 3: The six sampled camera trajectories, chosen to maximize viewpoint coverage of the input scene.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ transforms the multi-view latents synthesized by the video model $\mathcal{V}$ into an explicit 3D representation renderable from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) for this representation because it satisfies two requirements at once: it is explicit, and it renders in real time. The decoder architecture is detailed in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Prior feed-forward reconstructors regress one 3D Gaussian per input pixel and are therefore bounded by the number and resolution of views they can ingest: GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 , and AnySplat (Jiang et al., 2025a) trains on 24 images at 448×448 . Our video model, by contrast, synthesizes $V \times L =$

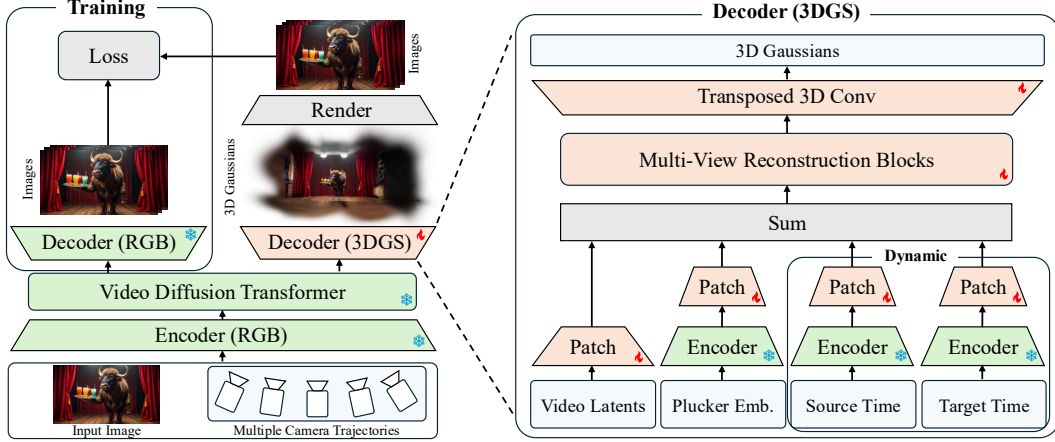


Figure 4: **3D generative reconstruction framework.** Our pipeline builds on a camera-controlled video diffusion model (Ren et al., 2025) pre-trained at large scale. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D renderings of the generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model, updating only the 3DGS decoder while keeping the pre-trained autoencoder and diffusion model frozen. At inference, the 3DGS decoder alone is used; the RGB decoder is no longer required. Time conditioning within the 3DGS decoder is precisely what lets the same architecture extend from static to dynamic 3D scene generation.

$6 \times 121 = 726$ input views at 704×1280 —more than an order of magnitude beyond these budgets, and well past the point at which pixel-space attention exhausts GPU memory. The bottleneck is intrinsic to pixel-space processing, in which attention over visual tokens scales in both memory and compute with the number of pixels. We sidestep it by declining to scale in pixel space at all, operating instead directly in the compressed latent space produced by the camera-controlled video diffusion model. We denote the resulting multi-view video latents by $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ projects its inputs into the hidden dimension of the main reconstruction blocks and predicts per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and the encoded Plücker embeddings $\mathbf{E}$, both of dimension $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input is patchified to the block hidden dimension; crucially, no additional visual encoder is required—a single spatial 2×2 patchification layer (Dosovitskiy et al., 2021) suffices to convert the video latents into flattened tokens. The elementwise sum of the two patchified inputs is then processed by the reconstruction blocks. Following Long-LRM (Ziwen et al., 2024), each block comprises one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers; we stack two such blocks, for 16 layers at 512 hidden dimensions. A final transposed 3D convolution maps the hidden representation to the 14 Gaussian channels—3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and color (r, g, b) . Compactly, the static decoder realizes the mapping $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. We first compute raw Plücker embeddings from the camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$, then encode them into the latent space by reusing the RGB encoder $\mathcal{E}$ of the pre-trained VAE. Because $\mathcal{E}$ expects three channels whereas a Plücker embedding has six, we encode its two three-dimensional components—the ray directions and the cross product of ray directions with origins—separately, and concatenate the results along the channel axis to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP then halves the channel dimension, mapping $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$ so that it aligns with the video latent $\mathbf{Z}$. Encoding cameras through the same VAE places them in exactly the latent space the reconstruction blocks operate in, which is what permits their direct summation with $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. The primary supervision is an image-space reconstruction objective on the student’s renderings, decomposed into a mean-squared-error term $\mathcal{L}_{mse}$ and a perceptual LPIPS term $\mathcal{L}_{lpiips}$ (Zhang et al., 2018) computed over VGG (Simonyan & Zisserman, 2014) features.

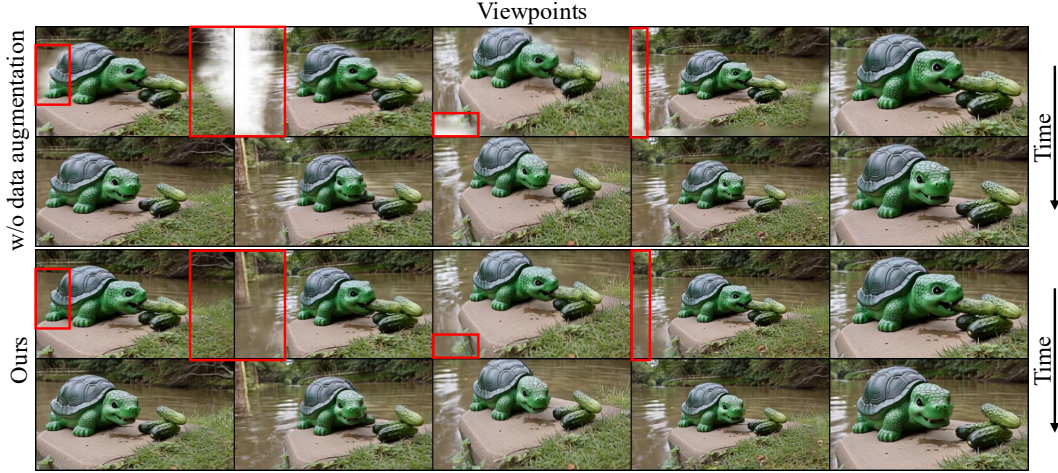


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder introduces artifacts in the generated 3D Gaussians: at early timesteps viewed from extreme poses, regions left uncovered by the supervision signal collapse to low opacity. Augmenting the supervision with a motion-reversed video guarantees that every timestep is observed with full spatial coverage, eliminating these low-opacity artifacts at the early timesteps.

The two terms are complementary: the former enforces pixel accuracy, the latter perceptual and high-frequency fidelity.

Depth supervision. Under the image loss alone, the recovered geometry tends to collapse onto flat surfaces. To counteract this, we additionally supervise the rendered depth maps against temporally consistent video depth from the off-the-shelf ViPE (Huang et al., 2025) estimator, using the scale-invariant depth loss $\mathcal{L}_{depth}$ of Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we impose an L_1 regularizer on opacity, $\mathcal{L}_{opacity}$, and prune the Gaussians occupying the lowest 80% of opacity, yielding a substantially more compact representation.

Total loss. The overall objective is the weighted sum of the four terms, with weights λ_i ,

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

A key strength of our framework is that it generalizes from static to dynamic 3D scenes with only minimal modifications, detailed below.

Self-distillation for dynamic scenes. The dynamic setup mirrors its static counterpart closely. In place of a single image, the video model now ingests a single video of length L together with its corresponding camera poses, and generates multi-view video latents that capture the same underlying motion. The data pipeline likewise parallels the static case: we sample diverse text prompts with large language models and synthesize videos with video diffusion models (Cosmos, 2025; Wan, 2025), then annotate the generated videos with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. Adopting the bullet-time formulation of Liang et al. (2025b), we design the dynamic 3DGS decoder $\mathcal{D}_s$ to predict time-dependent 3D Gaussians for every video frame. Architecturally, $\mathcal{D}_s$ reuses $\mathcal{D}_s$ almost verbatim; the sole addition is a pair of encoded temporal conditioning signals, the source and target time embeddings $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times, one per input frame, form $\mathbf{T}^{src, raw} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the raw target time is $\mathbf{T}^{tgt, raw} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially broadcast and augmented with a two-dimensional sinusoidal embedding to reach the three channels expected by $\mathcal{E}$, then encoded to the latent dimensions $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, with the target time replicated along the latent temporal axis to match

L' . At each training step we sample a target time uniformly at random, decode the corresponding 3D Gaussians, and supervise them across all trajectories produced by the dynamic data augmentation described below. We fine-tune $\mathcal{D}_d$ from the pre-trained static decoder $\mathcal{D}_s$, initializing the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zero so that optimization begins exactly at the static solution and departs from it only as the temporal signals become informative; the two embeddings are then added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder computes $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In static scenes, where time plays no role, frames from any timestep may supervise the 3DGS renderings. In the dynamic setting this no longer holds: because the Gaussians evolve over time, only frames sharing the matching timestamp constitute valid supervision. This tighter constraint admits a degenerate solution in which the Gaussians at a given timestep simply disregard information from frames at other timesteps (see Fig. 5), producing low-opacity artifacts in undersupervised regions.

We resolve this failure mode with a dynamic data augmentation strategy that rebalances supervision across timesteps. The imbalance is structural: early frames correspond to viewpoints near the input image, whereas later frames correspond to more distant ones. To furnish each timestep with both a near and a far observation, we reverse the input video in frame order and pass it through the video model $\mathcal{V}$, obtaining six additional multi-view sequences. Reversing these sequences back to the original motion order yields six inward-progressing trajectories—from far viewpoints toward the input—which, combined with the six original outward trajectories, provide 12 supervision views per timestep. This augmentation is confined to training: the reversed trajectories serve solely as additional supervision that forestalls the collapse induced by opacity pruning, and are not required at inference.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. Training uses no pre-existing multi-view dataset of any kind; instead, we construct the *Lyra* dataset, in which the video model alone supplies the supervision for the 3DGS decoder. The 3D setup comprises 59,031 images and the 4D setup 7,378 videos, all sourced from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both photorealistic and imaginative content. Synthesizing six camera trajectories per image (3D) or video (4D) expands these into 354,186 videos for 3D and 44,268 videos for 4D—a scale and diversity attainable precisely because the data are generated rather than captured.

Baselines. We benchmark against representative state-of-the-art methods for generative single-image-to-3D reconstruction: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is publicly available* for these methods, we cannot rerun them on our out-of-distribution evaluation set; we therefore compare against the quantitative results reported in their respective papers under the shared protocol described below.

Evaluation. Consistent with prior work, we evaluate single-image-to-3D on three standard benchmarks—RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks-and-Temples (Knapitsch et al., 2017)—following the evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b) and reporting the standard reconstruction metrics PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports the single-image-to-3D comparison. Our method attains the best PSNR, SSIM, and LPIPS on all three benchmarks, surpassing every prior approach across all nine metric-dataset combinations; the margin is most pronounced on DL3DV and Tanks-and-Temples, where it improves PSNR over the strongest available baseline by 3.45 and 3.34 dB, respectively. This consistency across benchmarks with markedly different content—from indoor real-estate scenes to unbounded outdoor captures—indicates that the gains stem from the framework rather than from tuning to any single distribution. A further consequence of the design is that the attainable quality is set by the teacher: because reconstructions are distilled from the video model, they should improve as

Table 1: **State-of-the-art comparison for single-image-to-3D generation** on RealEstate10K, DL3DV, and Tanks-and-Temples. Across all three datasets and all three metrics (PSNR, SSIM, LPIPS), our method attains the best score in every one of the nine metric–dataset cells; bold marks the best result in each column. Entries shown as “-” were not reported for the corresponding baseline.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS generation.** Five novel views rendered from 3DGS scenes generated by *Lyra*, exhibiting consistent geometry and appearance across wide viewpoint changes.

stronger camera-controlled video generators become available, without any change to our framework. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Fig. 1 and Fig. 6 present novel-view renderings from our generated 3DGS scenes (video results will be released upon acceptance). In both the static and dynamic settings, the model synthesizes high-quality content in previously unobserved regions while remaining faithful to the input image or video, corroborating the quantitative trends above. Qualitative comparisons on the *Lyra* dataset are deferred to Appendix C.

6.3 ABLATIONS

Tab. 2 isolates the contribution of each design choice by ablating key components on diverse out-of-distribution prompts, evaluating the 3DGS renderings against the videos generated by our camera-controlled video diffusion model. Read together, the ablations show that every component contributes and that the self-distillation data is the single most decisive factor. We report each variant relative to the full model (top row, 24.77 PSNR).

Real data only. Replacing self-distillation with training on real multi-view datasets alone—RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), the standard choices in prior work—collapses performance on out-of-distribution scenes, from 24.77 to 19.08 PSNR. This 5.69 dB gap, the largest of any ablation, directly quantifies the generalization advantage of learning from diverse generated supervision rather than from a fixed captured distribution.

Self-distillation + real data. Augmenting self-distillation with real data yields no improvement over self-distillation alone (24.74 vs. 24.77 PSNR). Far from a null result, this is an informative positive finding: the generated supervision

Table 2: **Ablation study on the *Lyra* dataset.** Removing any single component degrades performance relative to the full model (top row); replacing self-distillation with real data alone is the most damaging (−5.69 dB PSNR), and operating in pixel space rather than latent space is infeasible (OOM). Metrics are PSNR, SSIM, and LPIPS.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224



Figure 7: **Ablations.** Each variant is rendered from the same extreme novel viewpoint after image-to-3DGS generation, rendering the effect of removing each component directly comparable; depth visualizations and further ablations are provided in Appendix C.4.

is already sufficiently diverse and geometrically consistent to train the reconstruction model on its own, so real captures contribute nothing measurable and can be dispensed with entirely.

No depth loss. Removing the depth loss both flattens the recovered geometry and lowers the image-based metrics (24.31 vs. 24.77 PSNR); depth supervision thus improves geometry *and* appearance at once. Corresponding depth maps are visualized in Appendix C.4.

No opacity pruning. Retaining all Gaussians rather than pruning by opacity slightly degrades visual quality while leaving the representation far less compact. Because pruning removes the majority of Gaussians, it also accelerates rendering appreciably: at $H = 704$, $W = 1280$, a frame takes 18 ms with pruning versus 30 ms without—a $1.67\times$ speedup—so pruning improves quality and efficiency in tandem.

No LPIPS loss. The LPIPS loss confers robustness to input inconsistencies and preserves high-frequency detail; removing it sharply degrades perceptual quality, raising LPIPS from 0.224 to 0.370.

No multi-view fusion. The naive alternative—decoding 3D Gaussians for each trajectory independently and concatenating them into a single point cloud—performs markedly worse (17.73 vs. 24.77 PSNR). Learning the fusion jointly within our reconstruction blocks, where every token attends to all others, is therefore essential to the design rather than incidental.

No Mamba-2. Substituting Transformer-only blocks for the joint Transformer/Mamba-2 blocks slightly lowers quality while being dramatically slower: a single forward pass at $V = 6$, $L = 121$, $H = 704$, $W = 1280$ takes 3213 ms with the joint blocks against 20922 ms for Transformer-only—a $6.5\times$ speedup. The joint design thus advances quality and efficiency simultaneously.

No latent-based 3DGS. Operating the 3DGS decoder in pixel space rather than latent space simply exhausts memory: pixel-space methods such as BTimer (Liang et al., 2025b) are capped at 12 input frames, whereas our latent-space decoder ingests up to 726. Latent-space operation is what renders reconstruction at this scale feasible at all.

7 CONCLUSION

We presented *Lyra*, a unified framework for 3D and 4D scene generation from a single image or video. Rather than collecting multi-view datasets, we cast a camera-controlled video diffusion model as the teacher in a self-distillation framework that trains a student 3DGS decoder operating directly in the video model’s latent space and reconstructing 3D Gaussians with no post-processing or per-scene optimization. This design is what underlies the strong generalization and broad scene coverage we observe, attaining state-of-the-art single-image-to-3D reconstruction across three standard benchmarks; its dynamic extension, in turn, establishes the feasibility of feed-forward 4D generation from monocular video—a task not previously addressed—and substantially outperforms a strong feed-forward baseline. By construction, the scale and consistency of our scenes are bounded by the

capacity of the underlying video diffusion model. This dependence is a genuine limitation, since biases and failure modes of the teacher can propagate to the student; it is equally an asset, however, in that advances in camera-controlled video models translate directly into higher-quality reconstruction with no change to our framework. Promising avenues include adapting auto-regressive generation techniques (Chen et al., 2024a) to extend our scenes to larger spatial scales and, following concurrent work (Lin et al., 2025), modeling motion and tracking within the reconstruction network to further improve visual motion quality.

8 ETHICS STATEMENT

As with all generative AI technologies, our method carries a risk of misuse, notably the production of misleading or synthetic 3D/4D content. We take this risk seriously, yet the central purpose of this research is constructive: to advance simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. We anticipate that the work’s principal benefits will follow from the capabilities and insights below.

- **Realistic simulation for embodied agents:** The framework produces controllable, geometrically consistent camera motion within generated 3D and 4D environments, offering a substrate for training and evaluating agents that must perceive and navigate complex scenes.
- **Scalable data engines:** By removing the need for multi-view capture and per-scene optimization, our method produces high-fidelity, customizable scenes at scale—an approach well suited to research in robotics, reinforcement learning, and closed-loop simulation.
- **Improved technical understanding:** As an academic study, this work contributes a case study of how camera control interacts with spatiotemporal generative models, an understanding relevant to computer vision, graphics, and embodied AI.

While we caution against deceptive or otherwise wrongful use of this technology—where realistic generative outputs could yield misleading or unverifiable media—we hold that appropriate safeguards, such as provenance tracking, dataset documentation, and rigorous evaluation practices, can help ensure that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To enable full reproduction of our results and to accelerate future research in this area, we will release our training and inference code, model weights, and data for both 3D and 4D generation upon acceptance. Architectural specifications, the progressive training schedule, and evaluation protocols are documented in the appendix.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section elaborates on the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1 and details a refinement that further strengthens its 3D consistency.

Conservative mask refinement. A notable limitation of GEN3C’s forward warping is background leakage: occluded parts of an object in the source view (e.g., the lion’s body in Fig. 8) become visible from novel viewpoints yet go unmarked in the disocclusion mask. Consequently, the video diffusion model fails to complete these foreground regions, as seen in Fig. 8 (second and fourth columns).

We remedy this with a conservative masking strategy. Because the geometry of the occluded regions of $\mathbf{P}^{t,v}$ is unknown, we treat all such regions as occupied and derive refined disocclusion masks under this assumption, thereby prompting the video model to reason about the potentially visible foreground that standard point-based rendering misses.

Concretely, we construct a triangular mesh from the camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, following Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh acts as a boundary that separates visible from invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ against the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained by ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ is the original disocclusion mask from forward warping and ϵ is a small tolerance. Wherever the mesh surface lies closer than the original points, we conservatively mask the region as a potential foreground disocclusion. The result is substantially more reliable guidance for video generation, particularly under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

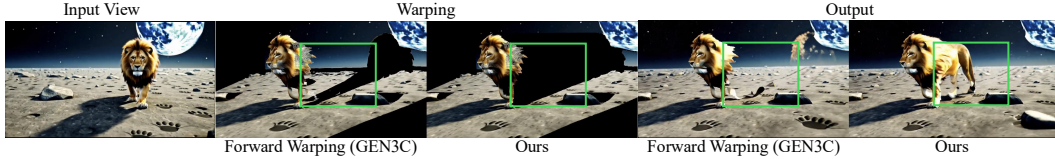


Figure 8: **Rendering function comparison.** Our model builds on GEN3C (Ren et al., 2025), which conditions camera control on forward-warped images. We contrast forward warping (left) with our conservative rendering function (right): under forward warping the highlighted region is erroneously filled with background pixels, whereas our approach correctly masks it, enabling the video model to complete it properly.

B MODEL DETAILS

Progressive training. Training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is prohibitively expensive, so we adopt a progressive schedule that grows resolution, sequence length, and view count in stages, as summarized in Tab. 3, and initialize the dynamic model from the final static checkpoint. Training completes in 6 days on 8 NVIDIA A100 80 GB GPUs, using gsplat (Ye et al., 2025) for the 3DGS implementation.

Subsampling Gaussians. The number of pixel-aligned 3D Gaussians grows linearly in both spatial and temporal resolution: our 726 frames at 704×1280 would entail 654,213,120 per-pixel Gaussians. We therefore emit a single 3D Gaussian per 8×8 spatial neighborhood—a $64\times$ reduction, to 10,222,080—after which opacity-based pruning shrinks the count further to 2,044,416, an additional $5\times$, yielding a far more compact representation without sacrificing quality.

Joint Transformer-Mamba blocks. We adopt the reconstruction block of Long-LRM (Ziwen et al., 2024): one Transformer layer followed by seven Mamba-2 layers, with two blocks stacked

Table 3: **Progressive training setup.** Per-stage configuration, listing resolution ($H \times W$), sequence length L , number of views V , supervision views S , batch size B , and optimization steps. Training progresses from low-resolution, single-view static stages to full-resolution multi-view static training, and finally to the dynamic stage, which is initialized from the last static checkpoint.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

to give 16 layers at 512 hidden dimensions. This network is considerably smaller than those of Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which use 24 layers at 1024 hidden dimensions. Although pure Transformers are generally more expressive, they are substantially slower to train and evaluate than Mamba-2; because we trained both configurations under an identical budget, we attribute the residual quality gap to insufficient iterations and expect Transformer-only blocks to require considerably more training to close it.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we encode cameras with Plücker embeddings, computed at full spatial and temporal resolution and then projected into the video latent space by the pre-trained RGB encoder. Because the encoder expects three input channels whereas a Plücker embedding is six-dimensional, we encode its two three-dimensional components—the ray directions and the cross product of ray directions with origins—separately and concatenate the resulting latents along the channel dimension.

Time encoding. Time is encoded by an analogous construction. To render the one-dimensional time value compatible with the encoder’s three-channel input, we concatenate a two-dimensional sinusoidal embedding along the channel dimension, replicate the result across the spatial dimensions, and encode it into the video latent space. This construction lets us encode the source and target times independently within the same compressed latent representation.

Dynamic data augmentation. Fig. 9 illustrates the dynamic data augmentation procedure on two example camera trajectories. For each motion state, the procedure yields a pair of supervision views that observe the scene from opposing extreme viewpoints, thereby balancing spatial coverage across timesteps.

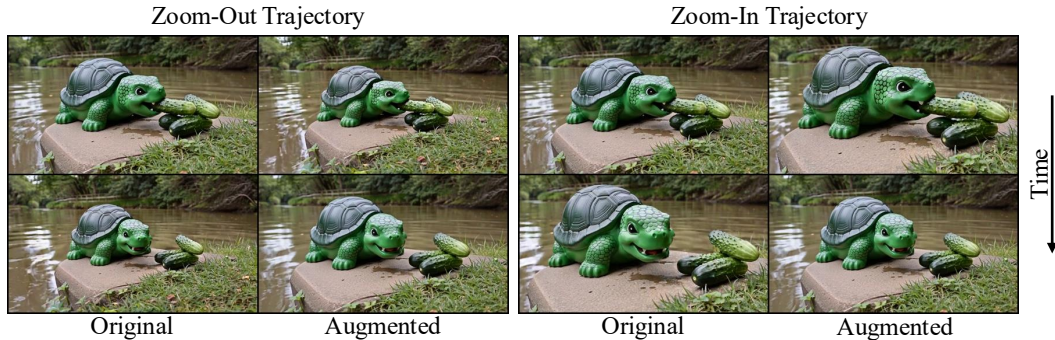


Figure 9: **Dynamic data augmentation videos.** We augment the supervision with a motion-reversed video so that every timestep is observed with full spatial coverage, forestalling low-opacity artifacts at the early timesteps. We show two example trajectories—zoom-out and zoom-in—together with their augmented counterparts, which reverse the camera motion.



Figure 10: **Image-to-3DGS comparison.** Five views rendered from scenes generated by our method and by BTimer (GEN3C) on the *Lyra* dataset. Our renderings exhibit markedly fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We further compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. To ensure a fair comparison on identical scenes, we asked the BTimer authors to run experiments for us on the same scenes as *Lyra*. Because BTimer is purely regression-based, we pair it with our GEN3C (Ren et al., 2025) video diffusion backbone: we adopt the same camera trajectories as our model and feed the RGB-decoded GEN3C videos to BTimer as input. BTimer operates in pixel space and requires 12 input images; as noted in the main paper, such pixel-space models exhaust memory when given the full 726 high-resolution GEN3C frames, so we uniformly subsample. For static evaluation we select 12 frames that evenly cover the available viewpoints; for dynamic evaluation we take 11 frames spanning the viewpoint range together with the bullet-time frame of the motion being reconstructed, guaranteeing that the target motion state is always available as a reference. We denote this integrated baseline BTimer (GEN3C), i.e., BTimer combined with GEN3C. This construction pairs BTimer with the same generative supervision available to our model, isolating the effect of the reconstruction decoder itself.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 presents qualitative comparisons with BTimer (GEN3C), and Tab. 4a the corresponding quantitative results. On this generated static distribution, our method is markedly higher in fidelity and exhibits substantially fewer artifacts, improving PSNR from 16.32 to 24.92—a gain of 8.60 dB that reflects the benefit of fusing all 726 views rather than the 12 to which the pixel-space baseline is confined.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation we compare against BTimer (GEN3C) (Sec. C.1) on 100 out-of-distribution videos from the dynamic *Lyra* dataset, cropping to 512×512 to match the resolution at which BTimer was primarily trained and thereby keep the comparison fair. As reported in Tab. 4b, our method substantially outperforms BTimer (GEN3C), improving PSNR from 20.29 to 23.07 (+2.78 dB) with consistent gains in SSIM and LPIPS. We note that this dynamic evaluation is conducted on our generated distribution against the adapted feed-forward

Table 4: **Quantitative results on the static and dynamic *Lyra* datasets.** Against BTimer (GEN3C)—BTimer supplied with the same GEN3C generations as our model—our method wins on every metric in both the static (a) and dynamic (b) settings; best results are in bold. Both comparisons are conducted on our generated distribution.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) **Static *Lyra* dataset.** On the generated static distribution, our method improves PSNR over BTimer (GEN3C) by 8.60 dB, with corresponding gains in SSIM and LPIPS.

(b) **Dynamic *Lyra* dataset.** On the generated dynamic distribution, our method improves PSNR over BTimer (GEN3C) by 2.78 dB, with corresponding gains in SSIM and LPIPS.

baseline; it establishes the feasibility and relative strength of feed-forward 4D generation rather than benchmark-level supremacy on captured dynamic scenes.

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes the depth maps recovered from our generated 3DGS. Depth supervision prevents the flat geometry that arises under image-only training, and does so without any degradation of visual quality.



Figure 11: **Depth loss ablation.** 3DGS renderings paired with their corresponding depth maps. Adding depth supervision prevents the flat geometry produced under image-only training, and does so without any loss of visual quality.

D ROBOT SIMULATION

A primary motivation for building a diverse 3D generator is its capacity to supply simulation environments for autonomous agents, and we illustrate this potential with an end-to-end pipeline. We first generate a 3D scene from text with our model, export the 3D Gaussians as a .ply file, and convert it to .usdz using the export function of 3DGUT (Wu et al., 2025a). The resulting .usdz asset imports directly into the NVIDIA Isaac robot development platform as a physically based virtual environment, within which AI-based robots can in principle be trained and evaluated. We present this as a qualitative proof of concept rather than a quantitative study: the demonstration establishes that the generated 3DGS is compatible with a standard simulation toolchain, and we leave a systematic evaluation of metric scale, collision quality, and downstream task performance to future work. Early demo results will be released upon acceptance; Fig. 12 shows a screenshot of the interface.

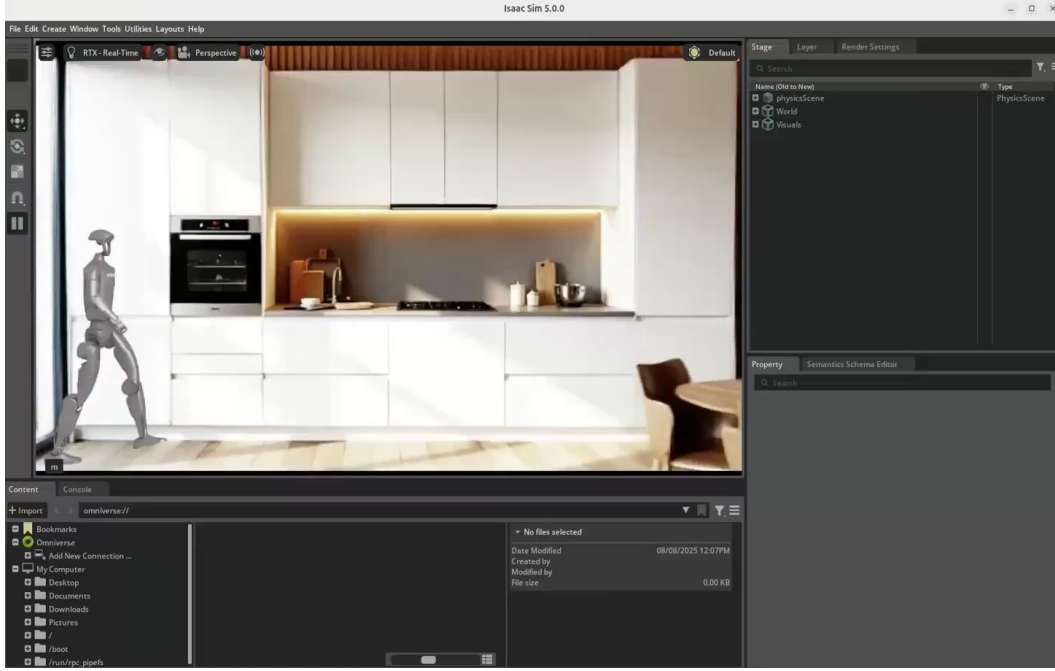


Figure 12: **Robot simulation.** A frame of a robot simulation running in the Isaac Sim 5.0 framework on a 3DGS scene generated by our method. This qualitative demonstration shows that our output imports into a standard simulation toolchain as a physically based environment; a quantitative study of metric scale and collision quality is left to future work.

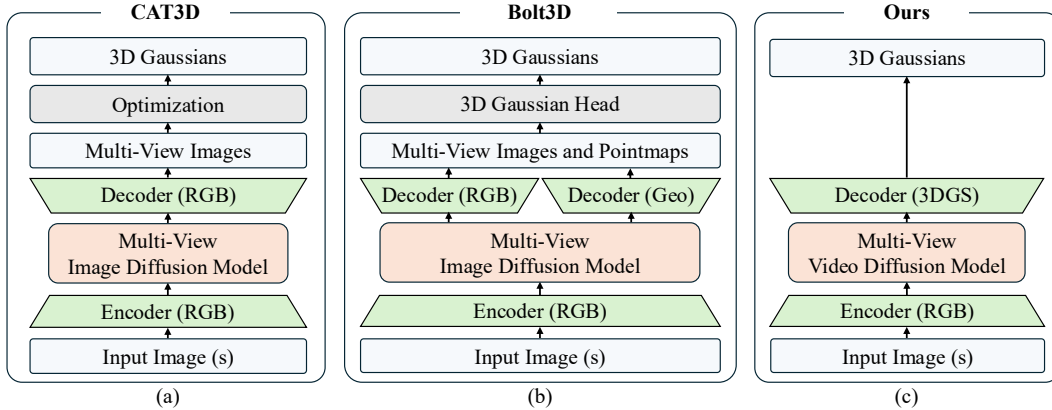


Figure 13: **Approaches to 3D generation.** CAT3D (Gao et al., 2024b) employs a multi-view image diffusion model to synthesize novel-viewpoint images, which a subsequent optimization stage reconstructs into 3D Gaussians. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to emit pointmaps via a geometry autoencoder and replaces optimization with a feed-forward 3D Gaussian head. Our work departs from both: building on a pre-trained camera-controlled video diffusion model, it *directly* decodes the multi-view video latents into 3D Gaussians in a single pass, with neither a separate optimization stage nor a geometry autoencoder.

E RELATED WORK

This section surveys additional related work and situates our method within it; Fig. 13 makes the distinctions from CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) explicit.

3D generation. Early 3D generation was largely confined to single object categories, extending GANs into 3D by exploiting neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). The advent of CLIP-based

supervision (Radford et al., 2021) broadened the field toward more flexible and diverse asset creation, unlocking both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). Diffusion models subsequently raised quality substantially by supplanting CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To reinforce structural coherence, one family of methods enforces multi-view consistency by generating a scene from multiple perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while another synthesizes scenes through iterative inpainting (Höllein et al., 2023; Shriram et al., 2024). A third direction lifts 2D images into 3D by coupling diffusion models with NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. A complementary body of work develops fast, feed-forward techniques that predict 3D content directly from image or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Unlike ours, however, these methods produce only static 3D scenes. Others specialize to particular content such as faces (Kirschstein et al., 2025), and concurrent work (Liang et al., 2025b; Xu et al., 2025) handles real scenes but neither generalizes to diverse generated scenes nor accommodates large viewpoint changes—precisely the regime our method targets.

4D generation. 4D generation—the synthesis of dynamic 3D scenes—has advanced rapidly, most often conditioned on text prompts or images. Following the introduction of large-scale generative models for the task (Singer et al., 2023), subsequent work has markedly improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, a substantial line of work lifts 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Related efforts inject physics priors into the generation pipeline (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), strengthen motion controllability through template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), or pursue compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

A further strand extends 3D GAN frameworks to the 4D setting by training directly on 2D video (Bahmani et al., 2023a; Xu et al., 2023). These methods are typically constrained by small-scale, single-category datasets that limit generalization and remain largely object-centric, neglecting background, so that their visual fidelity trails the photorealism of recent video generation models—the very fidelity our method inherits by distilling from such a model.

Camera-conditioned video models. Endowing video diffusion models with camera control has been a focus of recent progress. The pioneering MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) strengthened the conditioning by representing cameras as Plücker coordinates, whereas others follow MotionCtrl in simply flattening the poses (Wang et al., 2025b). A separate line (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) attains camera-motion control without

adding trainable parameters, and yet others adopt low-rank formulations (Zhang et al., 2025b). A common thread is that these approaches rely on U-Net-based architectures.

More recent work continues to extend U-Net-based camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024) while a growing body turns to diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited, and DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but relies on pre-defined, non-continuous camera trajectories. Others instead perform pose and geometry estimation with a video DiT (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b). CAT4D (Wu et al., 2025b) derives a multi-view video diffusion model by fine-tuning a multi-view backbone, and large-scale synthetic data further boosts quality, as SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrate with transformer-based architectures; 4DiM (Watson et al., 2024) instead trains a space-time diffusion model from scratch for single-image novel view synthesis. A further line edits the camera motion of videos with underlying motion via camera retargeting (Seo et al., 2025; Ma et al., 2025). Most relevant to us, a popular direction conditions camera-controlled video generation on a spatio-temporal cache built from, e.g., point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025); GEN3C, on which we build, belongs to this family.